

AI Audit Report — HW06 API Testing (SV 23127060 — Ninh Van Khai)

Phu luc bat buoc theo de bai muc 9.

Tuyen bo su dung AI

I use AI tools for the following tasks.

Toan bo qua trinh lam HW06 (sinh test case tu API spec, audit, mo rong, dung Postman collection,phan tich ket qua Newman, soan bao cao) deu co su tham gia cua cong cu AI. Moi luot tuong tac deu duoc ghi lai tu dong ngay tai thoi diem xay ra bang `agent-skill/eshop-api-23127060/scripts/ai_log.py`, khong viet lai tu tri nho.

Cong cu AI da dung	Vai tro
Claude Code (claude-opus-5)	Sinh / bien doi tai lieu va test case, chay script, tong hop bao cao

Cac ket qua so lieu (passed/failed) **khong** do AI uoc luong ma duoc tinh tu file `newman/*.json` that qua `scripts/summarize_newman.py`. So do bo sinh test do sinh vien **tu ve**, khong do AI sinh (de bai muc 11).

Sinh tu `ai/AI_log.md` luc 2026-09-01T16:04:02+07:00 · tong 14 luot tuong tac.

Phu luc A — Bang tuong tac AI

#	Thoi diem	Buoc	Tool	Noi dung	Prompt goc	Output
1	2026-09-01T14:29:39+07:00	STEP 0	Claude Code (claude-opus-5)	Trinh sat moi truong va doi chieu dac ta SUT	ai/prompts/step0_user_prompt.txt	ai/interactions/20260901sat-moi-truong-va-d
2	2026-09-01T14:36:00+07:00	STEP 1	Claude Code (claude-opus-5)	Lap dac ta may doc duoc + va 2 loi trong bo sinh test	ai/prompts/step1_spec.txt	ai/interactions/20260901dac-ta-may-doc-duoc-b_OUTPUT.md
3	2026-09-01T14:36:46+07:00	STEP 2a	Claude Code (claude-opus-5)	Vong 1/4 - sinh test case DOMAIN PARTITION	ai/prompts/step2a_dom.txt	ai/interactions/20260901sinh-test-case-doma
4	2026-09-01T14:37:10+07:00	STEP 2b	Claude Code (claude-opus-5)	Vong 2/4 - sinh test case STATE TRANSITION	ai/prompts/step2b_sta.txt	ai/interactions/20260901sinh-test-case-state

5	2026-09-01T14:37:58+07:00	STEP 2c	Claude Code (claude-opus-5)	Vong 3/4 - sinh test case SECURITY SEC-01..07	ai/prompts/step2c_sec.txt	ai/interactions/20260901-sinh-test-case-secu:
6	2026-09-01T14:37:58+07:00	STEP 2d	Claude Code (claude-opus-5)	Vong 4/4 - sinh test case SCHEMA VALIDATION	ai/prompts/step2d_sch.txt	ai/interactions/20260901-sinh-test-case-scher
7	2026-09-01T14:49:56+07:00	STEP 3	Claude Code (claude-opus-5)	Audit 225 test case bang bo luat tai lap duoc	ai/prompts/step3_audit.txt	ai/interactions/20260901-test-case-bang-bo-l1
8	2026-09-01T14:56:08+07:00	STEP 4	Claude Code (claude-opus-5)	Bo sung 18 test case AI bo sot, phan tich nguyen nhan	ai/prompts/step4_extend.txt	ai/interactions/20260901-sung-18-test-case-a:
9	2026-09-01T15:28:02+07:00	STEP 5	Claude Code (claude-opus-5)	Viet lai bo dung Postman collection, bo assertion gia	ai/prompts/step5_postman.txt	ai/interactions/20260901-lai-bo-dung-postman-
10	2026-09-01T15:28:02+07:00	STEP 6	Claude Code (claude-opus-5)	Chay Newman, phan tich that bai, chot moc hoi quy	ai/prompts/step6_newman.txt	ai/interactions/20260901-newman-phan-tich-tha:
11	2026-09-01T15:37:40+07:00	STEP 7	Claude Code (claude-opus-5)	Thu bang chung va viet bug report cho 34 bug	ai/prompts/step7_bugs.txt	ai/interactions/20260901-bang-chung-va-viet-l
12	2026-09-01T15:43:43+07:00	STEP 8	Claude Code (claude-opus-5)	Workflow CI/CD + kiem chung 2 lan chay tren may cuc bo	ai/prompts/step8_cicd.txt	ai/interactions/20260901-ci-cd-kiem-chung-2-
13	2026-09-01T15:50:21+07:00	STEP 9	Claude Code (claude-opus-5)	Thiet ke bo sinh test: pseudocode, mo ta so do, bao cao thiet ke	ai/prompts/step9_generator.txt	ai/interactions/20260901-ke-bo-sinh-test-psei
14	2026-09-01T16:04:02+07:00	STEP 10	Claude Code (claude-opus-5)	Bao cao chinh, AI Audit, AI Critique, xuất PDF, kiem tra bai nop	ai/prompts/step10_final.txt	ai/interactions/20260901-caoc-chinh-ai-audit-a:

Phu luc B — Chi tiet tung luot

#1 · Trình sat moi truong va doi chieu dac ta SUT

- Thoi diem: 2026-09-01T14:29:39+07:00 · Buoc: STEP 0 · Tool: Claude Code (claude-opus-5)
- Prompt: ai/prompts/step0_user_prompt.txt
- Output: ai/interactions/20260901T142939+0700_trinh-sat-moi-truong-va-doi-chieu-dac-ta_OUTPUT.md
- Files: report/00_environment.md, agent-skill/eshop-api-23127060/references/API_SPEC_NOTES.md, agent-skill/eshop-api-23127060/scripts/ai_log.py
- Human verified: pending

#2 · Lap dac ta may doc duoc + va 2 loi trong bo sinh test

- Thoi diem: 2026-09-01T14:36:00+07:00 · Buoc: STEP 1 · Tool: Claude Code (claude-opus-5)
- Prompt: ai/prompts/step1_spec.txt
- Output: ai/interactions/20260901T143600+0700_lap-dac-ta-may-doc-duoc-va-2-loi-trong-b_OUTPUT.md
- Files: spec/api-2.json, spec/_SCHEMA.md, agent-skill/eshop-api-23127060/scripts/gen_testcases.py, report/01_api_selection.md
- Human verified: pending

#3 · Vong 1/4 - sinh test case DOMAIN PARTITION

- Thoi diem: 2026-09-01T14:36:46+07:00 · Buoc: STEP 2a · Tool: Claude Code (claude-opus-5)
- Prompt: ai/prompts/step2a_dom.txt
- Output: ai/interactions/20260901T143646+0700_vong-1-4-sinh-test-case-domain-partition_OUTPUT.md
- Files: testcases/API-1_generated.csv, testcases/API-2_generated.csv, testcases/API-3_generated.csv
- Human verified: pending

#4 · Vong 2/4 - sinh test case STATE TRANSITION

- Thoi diem: 2026-09-01T14:37:10+07:00 · Buoc: STEP 2b · Tool: Claude Code (claude-opus-5)
- Prompt: ai/prompts/step2b_sta.txt
- Output: ai/interactions/20260901T143710+0700_vong-2-4-sinh-test-case-state-transition_OUTPUT.md
- Files: testcases/API-1_generated.csv, testcases/API-2_generated.csv, testcases/API-3_generated.csv
- Human verified: pending

#5 · Vong 3/4 - sinh test case SECURITY SEC-01..07

- Thoi diem: 2026-09-01T14:37:58+07:00 · Buoc: STEP 2c · Tool: Claude Code (claude-opus-5)
- Prompt: ai/prompts/step2c_sec.txt
- Output: ai/interactions/20260901T143758+0700_vong-3-4-sinh-test-case-security-sec-01-OUTPUT.md
- Files: testcases/API-1_generated.csv, testcases/API-2_generated.csv, testcases/API-3_generated.csv
- Human verified: pending

#6 · Vong 4/4 - sinh test case SCHEMA VALIDATION

- Thời điểm: 2026-09-01T14:37:58+07:00 · Bước: STEP 2d · Tool: Claude Code (claude-opus-5)
- Prompt: ai/prompts/step2d_sch.txt
- Output: ai/interactions/20260901T143758+0700_vong-4-4-sinh-test-case-schema-validation_OUTPUT.md
- Files: testcases/API-1_generated.csv, testcases/API-2_generated.csv, testcases/API-3_generated.csv
- Human verified: pending

#7 · Audit 225 test case bang bo luat tai lap duoc

- Thời điểm: 2026-09-01T14:49:56+07:00 · Bước: STEP 3 · Tool: Claude Code (claude-opus-5)
- Prompt: ai/prompts/step3_audit.txt
- Output: ai/interactions/20260901T144956+0700_audit-225-test-case-bang-bo-luat-tai-lap_OUTPUT.md
- Files: agent-skill/eshop-api-23127060/scripts/audit_testcases.py, testcases/API-1_audited.csv, testcases/API-2_audited.csv, testcases/API-3_audited.csv, report/03_audit.md, agent-skill/eshop-api-23127060/references/TESTCASE_TAXONOMY.md
- Human verified: pending

#8 · Bổ sung 18 test case AI bỏ sót, phân tích nguyên nhân

- Thời điểm: 2026-09-01T14:56:08+07:00 · Bước: STEP 4 · Tool: Claude Code (claude-opus-5)
- Prompt: ai/prompts/step4_extend.txt
- Output: ai/interactions/20260901T145608+0700_bo-sung-18-test-case-ai-bo-sot-phan-tich_OUTPUT.md
- Files: agent-skill/eshop-api-23127060/scripts/extend_testcases.py, testcases/API-1_final.csv, testcases/API-2_final.csv, testcases/API-3_final.csv, report/04_extend.md
- Human verified: pending

#9 · Viết lại bộ dung Postman collection, bỏ assertion già

- Thời điểm: 2026-09-01T15:28:02+07:00 · Bước: STEP 5 · Tool: Claude Code (claude-opus-5)
- Prompt: ai/prompts/step5_postman.txt
- Output: ai/interactions/20260901T152802+0700_viet-lai-bo-dung-postman-collection-bo-a_OUTPUT.md
- Files: agent-skill/eshop-api-23127060/scripts/build_collection.py, postman/collections/.json, postman/environments/.json, postman/scripts/schemas/.json, postman/data/.csv, report/05_postman_features.md
- Human verified: pending

#10 · Chạy Newman, phân tích thất bại, chốt môc hỏi quy

- Thời điểm: 2026-09-01T15:28:02+07:00 · Bước: STEP 6 · Tool: Claude Code (claude-opus-5)
- Prompt: ai/prompts/step6_newman.txt
- Output: ai/interactions/20260901T152802+0700_chay-newman-phan-tich-that-bai-chot-moc-hoi-quy_OUTPUT.md
- Files: newman/.html, newman/.json.gz, report/06_execution.md, ci/evidence/header_evidence.md, postman/contract_baseline/*.txt, agent-skill/eshop-api-23127060/scripts/summarize_newman.py, agent-skill/eshop-api-23127060/scripts/derive_contract.py, agent-skill/eshop-api-23127060/scripts/verify_header.py, agent-skill/eshop-api-23127060/scripts/run_newman.sh, agent-skill/eshop-api-23127060/scripts/run_datadriven.sh
- Human verified: pending

#11 · Thu bang chung va viet bug report cho 34 bug

- Thoi diem: 2026-09-01T15:37:40+07:00 · Buoc: STEP 7 · Tool: Claude Code (claude-opus-5)
- Prompt: ai/prompts/step7_bugs.txt
- Output: ai/interactions/20260901T153740+0700_thu-bang-chung-va-viet-bug-report-cho-34_OUTPUT.md
- Files: agent-skill/eshop-api-23127060/scripts/capture_bug_evidence.py, agent-skill/eshop-api-23127060/scripts/make_bug_report.py, bugs/BUG_REPORT.md, bugs/evidence/.md, bugs/ISSUE_TEMPLATES/.md
- Human verified: pending

#12 · Workflow CI/CD + kiem chung 2 lan chay tren may cuc bo

- Thoi diem: 2026-09-01T15:43:43+07:00 · Buoc: STEP 8 · Tool: Claude Code (claude-opus-5)
- Prompt: ai/prompts/step8_cicd.txt
- Output: ai/interactions/20260901T154343+0700_workflow-ci-cd-kiem-chung-2-lan-chay-tre_OUTPUT.md
- Files: .github/workflows/api-tests-23127060.yml, ci/api-tests-23127060.yml, ci/CI_CD_REPORT.md, ci/inject_failing_test.py, ci/evidence/local_ci_run_pass.log, ci/evidence/local_ci_run_fail.log, ci/evidence/header_evidence.md
- Human verified: pending

#13 · Thiet ke bo sinh test: pseudocode, mo ta so do, bao cao thiet ke

- Thoi diem: 2026-09-01T15:50:21+07:00 · Buoc: STEP 9 · Tool: Claude Code (claude-opus-5)
- Prompt: ai/prompts/step9_generator.txt
- Output: ai/interactions/20260901T155021+0700_thiet-ke-bo-sinh-test-pseudocode-mo-ta-s_OUTPUT.md
- Files: agent-skill/pseudocode/generator.pseudo.md, agent-skill/diagram/DIAGRAM_BRIEF.md, agent-skill/VIDEO_SCRIPT.md, report/07_test_generator_design.md
- Human verified: pending

#14 · Bao cao chinh, AI Audit, AI Critique, xuất PDF, kiem tra bai nop

- Thoi diem: 2026-09-01T16:04:02+07:00 · Buoc: STEP 10 · Tool: Claude Code (claude-opus-5)
- Prompt: ai/prompts/step10_final.txt
- Output: ai/interactions/20260901T160402+0700_bao-cau-chinh-ai-audit-ai-critique-xuat-OUTPUT.md
- Files: testcases/23127060_HW06_testcases.xlsx, ai/critique/AI_CRITIQUE.md, ai/audit/AI_AUDIT_REPORT.md, report/MAIN_REPORT.md, README.md, git-log/23127060_git_commit_log.txt, agent-skill/eshop-api-23127060/scripts/md_to_pdf.py, agent-skill/eshop-api-23127060/scripts/validate_submission.py
- Human verified: pending

Ghi chu

- Toan bo prompt goc (nguyen van) va output nam trong ai/interactions/.
- Cot Human verified = yes nghĩa là sinh viên đã đọc lại và chịu trách nhiệm về kết quả lượt đó.
- Số liệu passed/failed trong báo cáo được tính từ newman/*.json bằng scripts/summarize_newman.py, không do AI ước lượng.
- Số đo bộ sinh test (agent-skill/diagram/) do sinh viên tự vẽ, không do AI sinh.